

Assignment - I

Q) Prove that the strength of transverse fillet weld is 1.41 times the strength of parallel fillet weld

A) In case of shear stress in parallel fillet weld:-

$$\tau_{max} = \frac{F}{A_{min}} = \frac{F}{tL} = \frac{F}{\frac{h}{\sqrt{2}} \times L} \leq \hat{\tau}$$

$$F \leq \hat{\tau} \times \frac{h}{\sqrt{2}} L$$

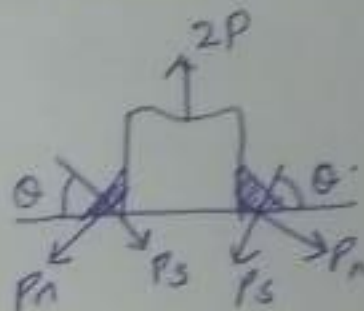
$$\boxed{F = \hat{\tau} \times \frac{h}{\sqrt{2}} L} \rightarrow (1)$$

In case of shear stress in transverse fillet weld:-

considering eq of vertical forces

$$2P = 2P_s \sin \theta + 2P_n \cos \theta$$

$$P = P_s \sin \theta + P_n \cos \theta$$



$\therefore$ the resultant of P_s & P_n is vertical, their horizontal components must be equal and opp.

$$\therefore P_s \cos \theta = P_n \sin \theta$$

$$P_n = \frac{P_s \cos \theta}{\sin \theta}$$

$$P = P_s \sin \theta + \frac{P_s \cos \theta \cos \theta}{\sin \theta}$$

Multiplying both sides of the above eqⁿ by $\sin \theta$

$$P \sin \theta = P_s \sin^2 \theta + P_s \cos^2 \theta$$

$$= P_s (\sin^2 \theta + \cos^2 \theta)$$

or

$$P_s = P \sin \theta$$

the width t' of the plane in the weld is inclined at an angle θ with horizontal

$$t' = \frac{h}{\sin \theta}$$

$$\tau = \frac{P_s}{h \cdot l}$$

$$\therefore \tau = \frac{P \sin \theta (\sin \theta + \cos \theta)}{h \cdot l}$$

In order to find out the plane with max. shear stress, differentiate τ w.r.t θ and set derivative equal to zero

$$\frac{d\tau}{d\theta} = 0$$

$$\frac{P}{h \cdot l} \frac{d}{d\theta} [\sin \theta (\sin \theta + \cos \theta)] = 0$$

$$\frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$x = \theta, u = \sin \theta, v = (\sin \theta + \cos \theta)$$

$$\frac{du}{d\theta} = \cos \theta, \frac{dv}{d\theta} = (\cos \theta - \sin \theta)$$

Substituting

$$\frac{d}{d\theta} (\sin \theta (\sin \theta + \cos \theta))$$

$$= \sin 2\theta + \cos 2\theta$$

$$\therefore \sin 2\theta + \cos 2\theta = 0$$

$$\tan 2\theta = -1$$

$$2\theta = 135^\circ$$

$$\theta = 67.5^\circ$$

$$\tau_{\max} = \frac{P \sin(67.5^\circ) [\sin(67.5^\circ) + \cos(67.5^\circ)]}{h \cdot l}$$

$$\tau_{\max} = \frac{1.21 P}{h \cdot l}$$

$$P_{\text{all}} = \frac{h \tau_{\max}}{1.21}$$

$$P_{\text{all}} = 0.8284 h \tau_{\max}$$

$$\therefore \frac{P_{\text{all for transverse weld}}}{P_{\text{all for parallel weld}}} = \frac{0.8284 h \tau_{\max}}{0.707 h \tau_{\max}} = 1.17$$

Hence Proved

Assignment - 2

Design procedure for cotter joint:-

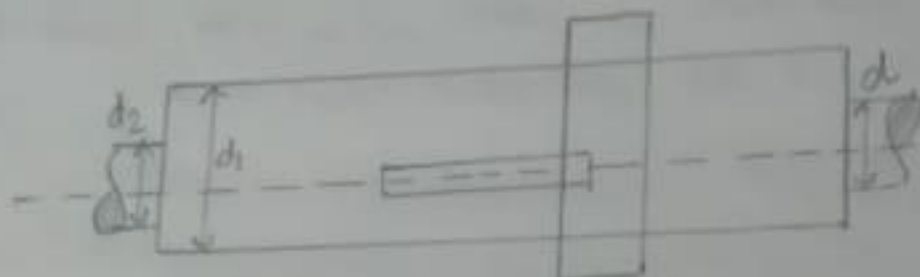


fig. cotter joint.

Let

P = tensile force acting on rods (N)

d = dia. of each rod (mm)

d_1 = outside diameter of socket (mm)

d_2 = diameter of spigot or inside dia. of socket (mm)

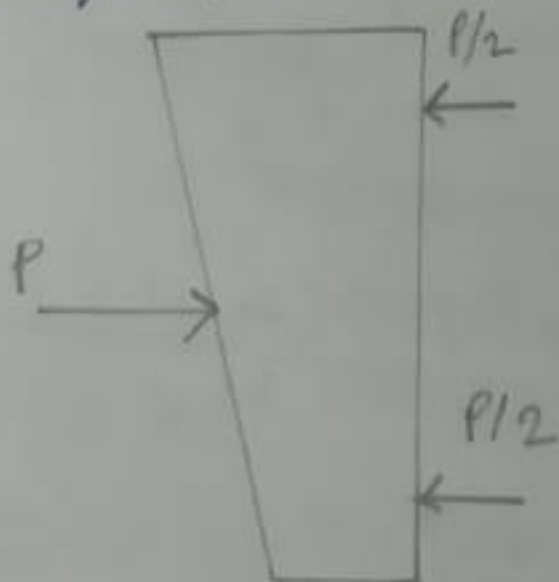
d_3 = dia. of spigot collar (mm)

d_4 = dia. of socket collar.

a = distance from end of slot to the end of spigot on rod-B (mm)

t = thickness of cotter (mm).

l = length of cotter (mm).



cotter.

i) Calculate the diameter of each rod:-

$$\sigma_t = \frac{P}{\left[\frac{\pi d^2}{4} \right]}$$

$$d = \sqrt{\frac{4P}{\pi \sigma_t}}$$

ii) Calculate the thickness of the collar by the empirical relationship given

$$t = 0.71d$$

iii) Calculate the diameter d_2 of the spigot on the basis of tensile stress:-

$$\sigma_t = \frac{P}{\frac{\pi}{4} d_2^2 - d_1 t}$$

$$P = \left[\frac{\pi}{4} d_2^2 - d_1 t \right] \sigma_t$$

iv) Calculate the outside diameter d_1 of the socket on the basis of tensile stress in the socket.

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2) t \right] \sigma_t$$

v) The diameter of spigot collar d_3 and the diameter of socket collar d_4 are calculated by following empirical relationships

$$d_3 = 1.5d$$

$$d_4 = 2.4d$$

vi) The dimensions a & c are calculated by the following empirical relationship

$$a = c = 0.75d$$

vii) Calculate the width b of the collar by shear consideration using eq. $P = 2b\tau z$ and bending consideration using eq. $P = 2ad_2$

and select the width, whichever is maximum between these two values.

$$b = \frac{P}{2\tau t} \quad \text{or} \quad b = \sqrt{\frac{3P}{\pi \sigma_b \left[\frac{d^2}{4} + \frac{d_4 - d_2}{6} \right]}}$$

viii) Check the crushing & shear stress in the spigot end.

$$\sigma_c = \frac{P}{td_2}, \quad \tau = \frac{P}{2ad_2}$$

ix) check the crushing & shear stresses in the socket end -

$$\sigma_c = \frac{P}{(d_1 - d_2)t}$$

$$\tau = \frac{P}{2(d_1 - d_2)c}$$

x) Cal. the thickness t_1 of the spigot collar by the following empirical relationship

$$t_1 = 0.45d_1$$

Ex: Two rods are connected by means of a collar joint. The inside diameter of the socket and outside diameter of socket collar are 50 and 100mm respectively. The rods are subjected to a tensile force of 50kN. The collar is made of steel 30C8 ($S_y = 400$) & factor of safety is 4. The width of collar is five times of thickness.

Calculate:-

- i) width & thickness of the collar on the basis of shear failure
- and
- ii) width & thickness of the collar on the basis of bending failure

Sol:

$$S_y = 400 \text{ N/mm}^2 \quad f(s) = 4$$

$$P = 50 \times 10^3 \text{ N} \quad d_1 = 100 \text{ mm}, \quad d_2 = 50 \text{ mm}$$

Step I:- Permissible stresses for collar

$$\sigma_t = \frac{S_{yt}}{f(s)} = \frac{400}{4} = 100 \text{ N/mm}^2$$

$$\tau = \frac{S_{sy}}{f(s)} = \frac{0.5 S_{yt}}{f(s)} = \frac{0.5 \times 400}{4} = 50 \text{ N/mm}^2$$

width & thickness on the basis of the shear failure -

$$b = 5t$$

$$P = 2b t \tau \quad \text{or} \quad 50 \times 10^3 = 2 \times 5t \times t \times 50$$

$$\therefore t = 10 \text{ mm} \quad \& \quad b = 5t = 50 \text{ mm}$$

width & thickness on the basis of bending failure:-

$$d_1 = 100 \text{ mm}, d_2 = 50 \text{ mm}$$

$$\sigma_b = \frac{\frac{f}{2} \left[\frac{d_2}{4} + \frac{d_1 - d_2}{2} \right] \frac{b}{2}}{\left(\frac{b^3 t}{12} \right)}$$

$$100 = \frac{\frac{50 \times 10^3}{2} \left[\frac{50}{4} + \frac{100 - 50}{2} \right] \frac{5t}{2}}{\frac{5b^3 t}{12}}$$

$$\therefore t = 0.27 \text{ or } 12 \text{ mm}$$

$$b = 5t = 60 \text{ mm}$$

DESIGN PROCEDURE FOR KNUCKLE JOINT:-

Let D = diameter of each rod (mm)

D_1 = enlarged diameter of each rod (mm)

d = diameter of knuckle pin (mm)

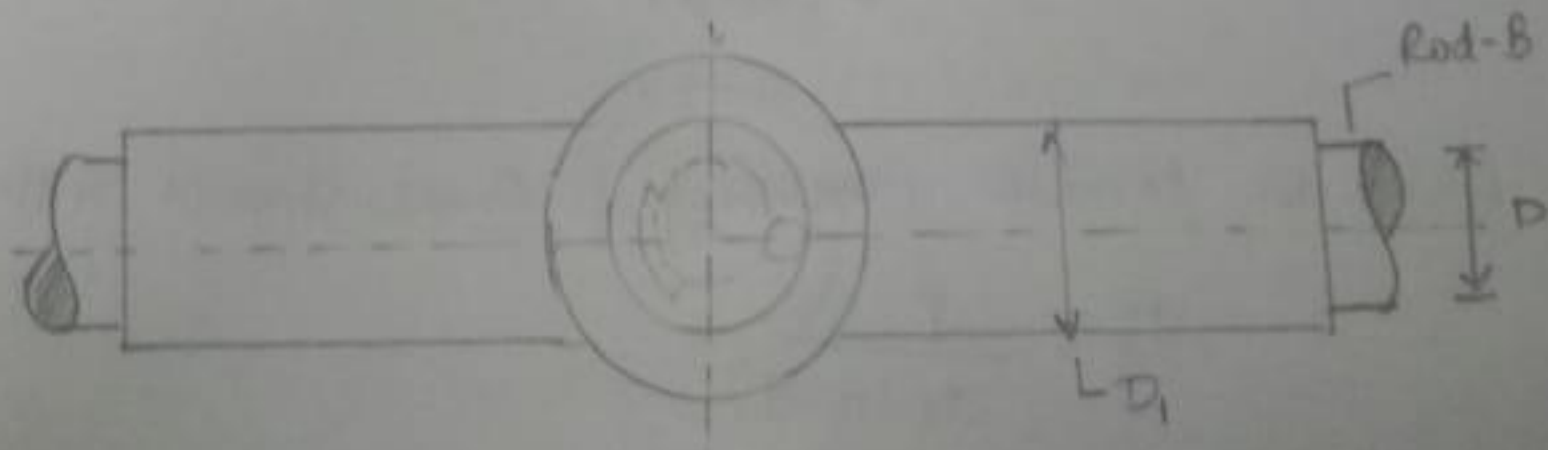
d_o = outside diameter of eye or fork (mm)

a = thickness of each eye of fork (mm)

b = thickness of eye end of rod-B (mm)

d_1 = diameter of pin head (mm)

x = distance of the centre of fork radius R from the eye (mm)



Knuckle joint

i) Calculate the diameter of each rod -

$$D = \sqrt{\frac{4P}{\pi \sigma_t}}$$

ii) Calculate the enlarged diameter of each rod by empirical relationship using $D_1 = 1.11 D$

iii) Calculate the dimensions a & b by empirical relationship using

$$a = 0.75 D \quad b = 1.25 D$$

iv) Calculate the diameters of the pin by shear consideration and bending consideration and select the diameter whichever is maximum.

$$d = \sqrt{\frac{2P}{\pi \tau}} = d \sqrt{\frac{32}{\pi \tau_b} \times \frac{P}{2} \left[\frac{L}{4} + \frac{a}{3} \right]}$$

v) Calculate the dimensions d_o & d_i by empirical relationships
 $d_o = 2d \quad d_i = 1.5d$

vi) Check the tensile, crushing & shear stresses in the eye

$$\sigma_t = \frac{P}{b(Pd_o - d)}$$

$$\sigma_c = \frac{P}{bd}$$

$$\tau = \frac{P}{b(d_o - d)}$$

vii) Check the tensile, crushing & shear stresses in the fork

$$\sigma_t = \frac{P}{2a(d_o - d)}$$

$$\sigma_c = \frac{P}{2ad}$$

Q: A wall-rack used to store round steel bars consists of two I-section cantilever beam fixed in the wall and the bars are stacked in a triangular fashion. The total wt. of bars is 75 kN. The permissible bending stress σ_b for the cantilevers is 165 N/mm^2 .



$$W = 75 \text{ kN}, \sigma_b = 165 \text{ N/mm}^2$$

$$M_b = 37.5 \times 10^3 \times \frac{2000}{3} = 25 \times 10^6 \text{ Nmm}$$

Calculation of I_{xx}

$$\frac{I_{xx}}{y} = \frac{M_b}{\sigma_b} = \frac{25 \times 10^6}{165} = 151.51 \times 10^3 \text{ mm}^3$$

Selection of beam

$$\frac{I_{xx}}{y} = \frac{1096.2 \times 10^4}{175/2} = 125.58 \times 10^3 \text{ mm}^3$$

Since the required I_{xx} is $151.55 \times 10^3 \text{ mm}^3$ beam is not suitable

Suppose beam (150 B 200) is suitable for the application

$$\frac{I_{xx}}{y} = \frac{1696.6 \times 10^4}{200/2} = 169.66 \times 10^3 \text{ mm}^3$$

$$\frac{I_{xx}}{y} > 151.51 \times 10^3 \text{ mm}^3$$